

Shot Quality Model

A logistic regression approach to assessing NHL shots on goal

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Abstract

The concept of examining NHL shot quality is not new; both Alan Ryder and Graeme Johns have published papers on the subject¹. Due to data limitations at the time of their papers, defense was examined team-wide; individual level data was available only for offense, whereas goalie data was not. Ryder's paper focused mainly on team defense and Johns wrote mainly about offense. Recently, data assigning each goaltender to the shot against became available. While mentioning the offensive side of the game, this paper will center on individual goalie (and some team level observations) defense. The conclusions reached here are very similar to those arrived at by both Johns and Ryder.

In order to assess NHL shot quality, data was collected from the 2003-04 regular season. A logistic regression model was built to predict the probability of a goal with four predictor variables—distance, shot type, rebound and situation. One minus this probability is the predicted probability of a save. The model fit the data rather well, with a c-statistic value of 0.733, and had a Kolmogorov-Smirnov (KS) statistic² of 35.05.

The model was validated against the 2002-03 shot data published by Alan Ryder. This data does not contain all 1,230 games, but has roughly 90% and is sufficient for an out-of-time model validation. KS, population stability and characteristics analyses were used to validate the model to this data.

For a given team, the predicted shooting percentages were compared to that of the average predicted as well as actual. Those above the league's average predicted value were taking higher quality shots, while those below lower. Comparisons to actual shooting percentage were also made.

For the flip side of the coin, the predicted save percentages for each goaltender were compared to that of the average predicted value. Those falling below were said to have faced "tougher" shots, those above "easier" shots and those at the mean, "average" shots. One could make inferences from this classification. The predicted save

¹ See *Shot Quality – A Method for the Study of the Quality of a Hockey Team's Shots Allowed* [Ryder, January 2004] and *Statistical Shot Quality Weighting – An Investigation of NHL Shot Quality and Offensive Performance* [Johns, October 2004]

² See Glossary for definition of KS.

percentages were also compared to actual save percentages and indexed to see who outperformed predictions and vice-versa. This also speaks to which teams faced higher percentage (i.e., higher quality) shots *versus* those that did not. Perhaps some teams played differently in front of their different goalies. This paper attempts to address these issues.

Data

The data used for the model was collected from the NHL play-by-play (PBP) files as well as game summary (GS) files—both available at www.nhl.com. A few PBP files stopped short of recording the entire game; the data does not tie out to the year-end statistics published by the NHL, but is very close and any differences are statistically immaterial.

The PBP files were generated by the Real Time Scoring System (RTSS), of which much has already been written. These files provide some information on each of the “events” occurring throughout a game and the shots on goal data was used for this study.

Shot Type	Goals			Saves			Total	
	#	Dist	Int Rate	#	Dist	Int Rate	#	Dist
Backhand	567	9.0%	11.2%	4,484	7.2%	88.8%	5,051	7.3%
Slap	1,645	26.0%	6.4%	23,969	38.2%	93.6%	25,614	37.1%
Snap	988	15.6%	9.7%	9,185	14.7%	90.3%	10,173	14.7%
Tip-In	755	12.0%	21.4%	2,770	4.4%	78.6%	3,525	5.1%
Wrap	54	0.9%	5.4%	941	1.5%	94.6%	995	1.4%
Wrist	2,308	36.5%	9.8%	21,328	34.0%	90.2%	23,636	34.3%
Total	6,317	100.0%	9.2%	62,677	100.0%	90.8%	68,994	100.0%

Table 1: Shot Type

As shown in Table 1 above, slap shots were employed the most, 37.1% of the time, with a success rate of 6.4% (accounting for 26.0% of all goals).

Methodology

A binary target variable called GOAL was created with a value of 1 for a goal and 0 for a save. The data was then randomly split 75%/25% for model training, or development, and validation. A logistic regression model was constructed on the 75% training sample and validated on the 25% holdout portion of the data. This was done to ensure the model does not over fit the data. The model was subsequently validated against an out-of-time dataset, namely that from the 2002-03 regular season. Johns also observed that

a logistic function fit the shot distance. The model herein considered several variables for inclusion:

The variables considered for the model included:

- Distance (measured in feet)
- Shot Type
- Situation
- Period
- Penalty Shot
- Rebound
- Scramble Shot
- Long Shot
- Normal Shot

Rebound, Scramble, Long and Normal shots were defined per Alan Ryder's paper³. True interaction terms were not considered here, but are inherent in the model since several variables are working simultaneously.

Model Results

Four variables (plus an intercept) remained in the final model—distance, shot type, rebound and situation; the others were either too correlated to these or insignificant.

³ See Glossary for definitions.

Variable	Range	Points
Intercept	Add to all records	-2.2369
Distance	Less than 10 ft	0.6884
	10 - 12 ft	0.6374
	13 - 14 ft	0.5564
	15 - 16 ft	0.5174
	17 - 22 ft	0.3654
	23 - 31 ft	0.0000
	32 - 36 ft	-0.3805
	37 - 38 ft	-0.4758
	39 - 44 ft	-0.8155
	45 - 57 ft	-1.0848
	58 ft or more	-1.3824
Shot Type	Wrap	-0.0742
	Slap	-0.0573
	Wrist	0.0093
	Snap	0.0130
	Backhand	0.0361
	Tip-In	0.1487
Rebound	Yes	1.3362
	No	0.0000
Situation	EV	-0.1244
	SH	0.0399
	PP	0.4007

Table 2: Model Scorecard

The model score, or predicted probability of a goal, is defined using Table 2 as follows:

$$P(\text{GOAL}) = \frac{1}{1 + e^{-\sum \text{points}}}.$$

The c-statistic for the model was 0.733. The c-statistic is the probability that an observation with an event (goal, in our case) has a higher *predicted* probability than an observation without an event. Values less than 0.50 mean the model does no better than randomly assigning a goal to a particular observation.

The KS statistic was used to validate the model. When comparing the training and validation datasets, we wish to see a small difference in KS values. Table 3 below illustrates this and we can safely say the model does not over fit (35.09 compared to 34.91).

KS Report for Training Sample									
Pred Probability (%)	Totals	Cuml %	Saves	Int Rate	Cuml %	Goals	Int Rate	Cuml %	KS
17.05 - 100.00	5,267	100.00%	3,912	74.27%	100.00%	1,355	25.73%	100.00%	0.00
14.22 - 17.04	5,207	89.81%	4,372	83.96%	91.67%	835	16.04%	71.20%	20.47
12.07 - 14.21	4,757	79.74%	4,156	87.37%	82.37%	601	12.63%	53.45%	28.92
8.70 - 12.06	5,185	70.54%	4,617	89.05%	73.53%	568	10.95%	40.68%	32.85
6.13 - 8.69	5,523	60.51%	5,076	91.91%	63.70%	447	8.09%	28.61%	35.09
4.85 - 6.12	4,158	49.82%	3,943	94.83%	52.90%	215	5.17%	19.11%	33.79
3.76 - 4.84	6,398	41.78%	6,118	95.62%	44.51%	280	4.38%	14.54%	29.97
2.93 - 3.75	4,560	29.40%	4,446	97.50%	31.49%	114	2.50%	8.59%	22.90
2.35 - 2.92	5,284	20.58%	5,106	96.63%	22.02%	178	3.37%	6.16%	15.86
0.00 - 2.34	5,355	10.36%	5,243	97.91%	11.16%	112	2.09%	2.38%	8.78
Total	51,694		46,989	90.90%		4,705	9.10%		35.09

KS Report for Validation Sample									
Pred Probability (%)	Totals	Cuml %	Saves	Int Rate	Cuml %	Goals	Int Rate	Cuml %	KS
17.16 - 100.00	1,641	100.00%	1,218	74.22%	100.00%	423	25.78%	100.00%	0.00
14.25 - 17.15	1,709	90.51%	1,441	84.32%	92.24%	268	15.68%	73.76%	18.48
12.07 - 14.24	1,796	80.64%	1,535	85.47%	83.05%	261	14.53%	57.13%	25.92
8.70 - 12.06	1,709	70.25%	1,512	88.47%	73.27%	197	11.53%	40.94%	32.32
6.13 - 8.69	1,870	60.38%	1,723	92.14%	63.63%	147	7.86%	28.72%	34.91
4.85 - 6.12	1,338	49.57%	1,256	93.87%	52.65%	82	6.13%	19.60%	33.04
3.76 - 4.84	2,136	41.83%	2,023	94.71%	44.64%	113	5.29%	14.52%	30.12
2.93 - 3.75	1,588	29.49%	1,551	97.67%	31.74%	37	2.33%	7.51%	24.24
2.35 - 2.92	1,781	20.31%	1,728	97.02%	21.86%	53	2.98%	5.21%	16.65
0.00 - 2.34	1,732	10.01%	1,701	98.21%	10.84%	31	1.79%	1.92%	8.92
Total	17,300		15,688	90.68%		1,612	9.32%		34.91

Table 3: KS Reports for Training and Validation Samples

The overall score, or predicted probability, distribution of the model is as follows:

KS Report for Entire Dataset									
Pred Probability (%)	Totals	Cuml %	Saves	Int Rate	Cuml %	Goals	Int Rate	Cuml %	KS
17.05 - 100.00	7,092	100.00%	5,278	74.42%	100.00%	1,814	25.58%	100.00%	0.00
14.22 - 17.04	6,958	89.72%	5,852	84.10%	91.58%	1,106	15.90%	71.28%	20.30
12.07 - 14.21	6,327	79.64%	5,504	86.99%	82.24%	823	13.01%	53.78%	28.47
8.70 - 12.06	6,894	70.47%	6,129	88.90%	73.46%	765	11.10%	40.75%	32.71
6.13 - 8.69	7,393	60.47%	6,799	91.97%	63.68%	594	8.03%	28.64%	35.05
4.85 - 6.12	5,496	49.76%	5,199	94.60%	52.83%	297	5.40%	19.23%	33.60
3.76 - 4.84	8,534	41.79%	8,141	95.39%	44.54%	393	4.61%	14.53%	30.01
2.93 - 3.75	6,148	29.42%	5,997	97.54%	31.55%	151	2.46%	8.31%	23.24
2.35 - 2.92	7,065	20.51%	6,834	96.73%	21.98%	231	3.27%	5.92%	16.06
0.00 - 2.34	7,087	10.27%	6,944	97.98%	11.08%	143	2.02%	2.26%	8.82
Total	68,994		62,677	90.84%		6,317	9.16%		35.05

Table 4: KS Report for Entire Dataset

Another method of assessing the model is a fit chart. A fit chart measures the average predicted probability per interval (see Table 4 above) against the actual goal rate in that interval. As shown in Illustration 1, despite some rank-ordering issues near the bottom of the distribution, the predicted values fit the actual values rather well.

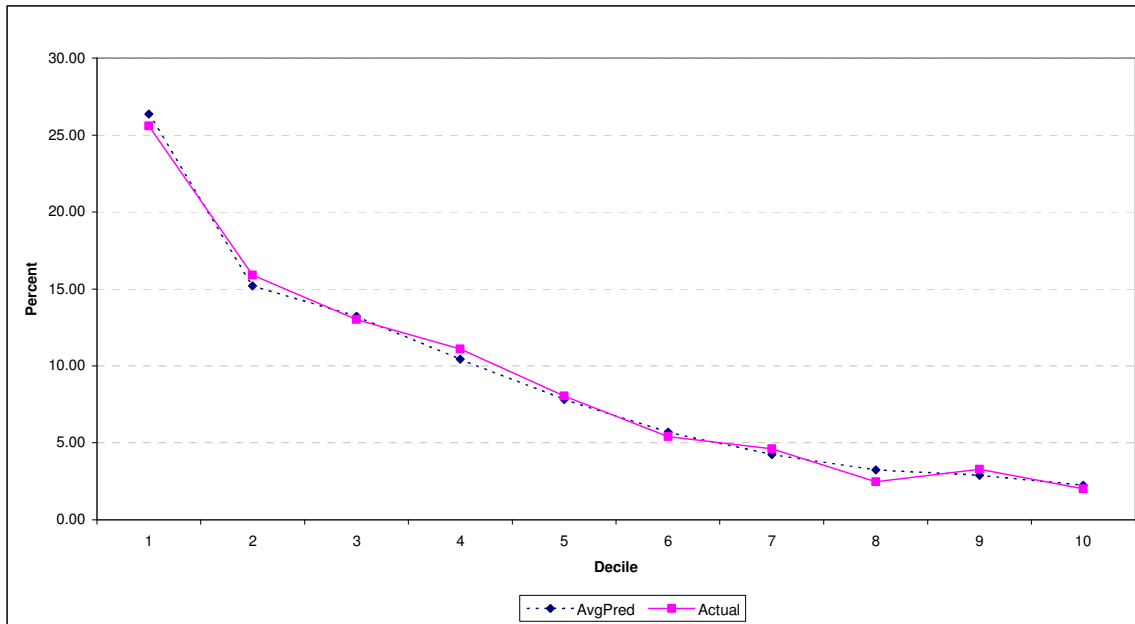


Illustration 1: Model Fit Chart

To further test the model, an out-of-time dataset was obtained from Alan Ryder. This data contains the shots on goal information from the PBP files for the 2002-03 regular season. It should be noted that the first 127 games are missing from the file. However, since we have around 90% of the games this should suffice for our purposes. Three measurements were run against this data—KS report, population stability index (PSI) and characteristics analysis⁴.

KS Report for Out of Time Validation 2002-03 Data										
Pred Probability (%)	Totals	Cuml %	Saves	Int Rate	Cuml %	Goals	Int Rate	Cuml %	KS	
17.16 - 100.00	6,172	100.00%	4,587	74.32%	100.00%	1,585	25.68%	100.00%	0.00	
14.22 - 17.15	6,750	90.14%	5,643	83.60%	91.92%	1,107	16.40%	72.69%	19.23	
12.07 - 14.21	5,840	79.35%	5,042	86.34%	81.98%	798	13.66%	53.61%	28.37	
8.70 - 12.06	6,328	70.02%	5,675	89.68%	73.10%	653	10.32%	39.86%	33.24	
6.14 - 8.69	6,062	59.91%	5,530	91.22%	63.11%	532	8.78%	28.61%	34.50	
4.85 - 6.13	5,713	50.22%	5,401	94.54%	53.37%	312	5.46%	19.44%	33.93	
3.76 - 4.84	7,424	41.09%	7,085	95.43%	43.86%	339	4.57%	14.06%	29.79	
2.93 - 3.75	5,283	29.23%	5,143	97.35%	31.38%	140	2.65%	8.22%	23.16	
2.35 - 2.92	6,667	20.79%	6,461	96.91%	22.32%	206	3.09%	5.81%	16.51	
0.00 - 2.34	6,342	10.13%	6,211	97.93%	10.94%	131	2.07%	2.26%	8.68	
Total	62,581		56,778	90.73%		5,803	9.27%		34.50	

Table 5: KS Report for Out-of-Time 2002-03 Data

⁴ See Glossary for definition of PSI and characteristics analysis.

As shown in Table 5, the KS of 34.50 is close to that of our baseline (2003-04 data) KS of 35.05. Some power was lost, but is minimal. The score, or predicted probability, distribution from the baseline data (2003-04) was compared to that of the validation period (2002-03) and the PSI was computed.

Pred Probability (%)	Baseline		Validation		(B) - (A)	(B) / (A)	ln(B / A)	(B - A) * ln(B/A)
	Number of Records	Percent of Total Records (A)	Number of Records	Percent of Total Records (B)	Distribution Difference	Difference in Proportion	Natural Log of Proportion Diff	Population Divergence
17.05-High	7,092	10.28%	6,721	10.74%	0.46%	104.48%	0.044	0.0002
14.22-17.04	6,958	10.08%	6,201	9.91%	-0.18%	98.25%	-0.018	0.0000
12.07-14.21	6,327	9.17%	5,840	9.33%	0.16%	101.76%	0.017	0.0000
8.70-12.06	6,894	9.99%	6,328	10.11%	0.12%	101.20%	0.012	0.0000
6.13-8.69	7,393	10.72%	6,741	10.77%	0.06%	100.52%	0.005	0.0000
4.85-6.12	5,496	7.97%	5,034	8.04%	0.08%	100.98%	0.010	0.0000
3.76-4.84	8,534	12.37%	7,424	11.86%	-0.51%	95.91%	-0.042	0.0002
2.93-3.75	6,148	8.91%	5,283	8.44%	-0.47%	94.74%	-0.054	0.0003
2.35-2.92	7,065	10.24%	6,667	10.65%	0.41%	104.04%	0.040	0.0002
0.00-2.34	7,087	10.27%	6,342	10.13%	-0.14%	98.66%	-0.014	0.0000
Total	68,994	100.00%	62,581	100.00%	Population Stability Index			0.0009

Table 6: Population Stability Index: Model Data Compared to 2002-03 Data

The very small PSI of 0.0009 indicates little divergence in population distributions, i.e., the model built shows stability when applied to an out-of-time sample. In addition, the distribution of each model characteristic was also compared between time periods.

Variable	Range	2003-04		2002-03		Percent Difference (A)	Model Point Values (B)	Weighted Change (A * B)
		Number in Population Standard	Percent in Population Standard	Number in Validation Sample	Percent in Validation Sample			
Shot Distance	Less than 10 ft	2,945	4.27%	2,568	4.10%	-0.17%	0.6884	-0.0011
	10 - 12 ft	4,974	7.21%	4,541	7.26%	0.05%	0.6374	0.0003
	13 - 14 ft	3,570	5.17%	3,297	5.27%	0.09%	0.5564	0.0005
	15 - 16 ft	3,040	4.41%	2,799	4.47%	0.07%	0.5174	0.0003
	17 - 22 ft	7,619	11.04%	7,033	11.24%	0.20%	0.3654	0.0007
	23 - 31 ft	9,095	13.18%	8,356	13.35%	0.17%	0.0000	0.0000
	32 - 36 ft	4,934	7.15%	4,500	7.19%	0.04%	-0.3805	-0.0001
	37 - 38 ft	2,150	3.12%	1,935	3.09%	-0.02%	-0.4758	0.0001
	39 - 44 ft	6,620	9.60%	5,812	9.29%	-0.31%	-0.8155	0.0025
	45 - 57 ft	15,078	21.85%	13,574	21.69%	-0.16%	-1.0848	0.0018
	58 ft or more	8,969	13.00%	8,166	13.05%	0.05%	-1.3824	-0.0007
		68,994	100.00%	62,581	100.00%			0.0043
Shot Type	Wrap	995	1.44%	860	1.37%	-0.07%	-0.0742	0.0001
	Slap	25,614	37.12%	24,104	38.52%	1.39%	-0.0573	-0.0008
	Wrist	23,636	34.26%	21,198	33.87%	-0.39%	0.0093	0.0000
	Snap	10,173	14.74%	8,594	13.73%	-1.01%	0.0130	-0.0001
	Backhand	5,051	7.32%	4,608	7.36%	0.04%	0.0361	0.0000
	Tip-In	3,525	5.11%	3,217	5.14%	0.03%	0.1487	0.0000
		68,994	100.00%	62,581	100.00%			-0.0009
Rebound	Yes	1,781	2.58%	2,174	3.47%	0.89%	1.3362	0.0119
	No	67,213	97.42%	60,407	96.53%	-0.89%	0.0000	0.0000
		68,994	100.00%	62,581	100.00%			0.0119
Situation	EV	53,624	77.72%	48,425	77.38%	-0.34%	-0.1244	0.0004
	SH	2,512	3.64%	2,380	3.80%	0.16%	0.0399	0.0001
	PP	12,858	18.64%	11,776	18.82%	0.18%	0.4007	0.0007
		68,994	100.00%	62,581	100.00%			0.0012
Total Deviation in Logit from Baseline								0.0166

Table 7: Characteristics Analysis: Model Data Compared to 2002-03 Data

Given the very low PSI value, this measurement was not really necessary. However, it was computed to explain any deviation in model score distribution from the baseline to the 2002-03 data. As seen in Table 7 above, there was very little difference in the distribution of these variables from one season to the next.

Offense Results

Now that we are satisfied with the model, let us examine shooting percentage for 2003-04. Using the above model, each shot was given a (predicted) probability of being a goal. The average predicted shooting percentage (excluding empty net goals) was 9.1, slightly higher than the actual of 8.9; that is, people scored at a lower rate than predicted, given the quality of shots taken. We can now compare each team's (and each individual's) shooting percentage with that of the entire league to determine which team had lower, higher or average shot quality.

Shot Quality	SHOT%	Number of Teams
Average	Predicted = 9.1	1
Lower	Predicted < 9.1	14
Higher	Predicted > 9.1	15

Table 8: Shot Quality Definition & Distribution

When examining team shots (non-empty net goals) using this definition of shot quality we see:

Shot Quality	Mean Predicted SHOT%	Standard Deviation	Lower Bound (95%)	Upper Bound (95%)
Average	9.1			
Lower	8.7	0.3	8.6	8.9
Higher	9.5	0.3	9.4	9.7

Table 9: 95% Confidence Intervals for Mean Predicted Shooting Percentage

While our definition of shot quality in Table 8 may appear rigid, it is statistically appropriate. Below are some observations and inferences on a team-wide basis (these can, of course, be extended to the individual level, but that is beyond the scope of this paper and have been covered by others).

Columbus:

The Blue Jackets had a much higher shot quality than the league average (9.9 compared to 9.1), but failed miserably to achieve projected results—actual shot percentage was 7.6. The chances, statistically, were there; they just couldn't capitalize on them.

Detroit:

With snipers such as Brett Hull on their team, it was no wonder that the Red Wings outperformed projections by a wide margin. On the whole, they took lower quality shots (8.4), but converted at 10.0. This speaks to the skill level on their roster, e.g., Hull, Datsyuk, et. al. Perhaps the fear of the Red Wings' stars influenced the play of their opponents' goalies as well?

New Jersey:

Considered a trapping, defense-first team, it was apt that the Devils sacrificed offensive shot quality (8.4, well below the league average). This goes well with the observation in the goalie results section further on in this paper. New Jersey did, however, perform better than anticipated (8.6 actual), though still finished below the league actual average of 8.9 percent.

Tampa Bay:

On the flip side of the defense-first moniker is the Lightning, thought to be a wide-open, offensive team. Their shot quality was slightly below average (9.0 vs. 9.1), but they scored better than anticipated, coming in at 9.6.

Toronto:

The Maple Leafs is another example of a team with approximately average predicted shot quality (9.0) that performed significantly better (10.3).

Goalie Results

Let us now examine the data from a goaltender's perspective. This can subsequently be extended to the team for which he plays. Recall, the model assigned each shot a (predicted) probability of being a goal. One minus this probability is, hence, the (predicted) probability of a shot being saved.

The average predicted save percentage for all goalies (excluding empty net goals, which go in every time) was .909. The actual, observed, save percentage for all goalies in 2003-04 was .911, somewhat higher than predicted. We can now compare each goalie's predicted save percentage to that of all goalies (.909) to determine if he faced "easier," "tougher" or "average" shot qualities (i.e., shots with lower, higher or average

predicted probabilities of going in). Two questions immediately come to mind when viewing these results: (1) is the goalie performing as predicted? and (2) is the team “allowing” higher percentage shots with one goalie in net *versus* a back-up?

In total, there were 100 goalie / team combinations studied. For example, Sean Burke was examined twice—his numbers in Philadelphia as well as Phoenix. The shot quality was defined and distributed as follows:

Shot Quality	SV%	Number of Goalies
Average	Predicted = .909	6
Easier	Predicted > .909	44
Tougher	Predicted < .909	50

Table 10: Shot Quality Definition & Distribution

One might think that this interpretation of shot quality may be strict; is a predicted save percentage of .908 or .910 really different from .909? Probably not in practice, but if we attach confidence intervals to the average predicted save percentage, we see that the classification is, statistically speaking, apt.

Shot Quality	Mean Predicted SV%	Standard Deviation	Lower Bound (95%)	Upper Bound (95%)
Average	.909			
Easier	.914	.007	.912	.916
Tougher	.901	.009	.898	.904

Table 11: 95% Confidence Intervals for Mean Predicted Save Percentage

Below are some inferences from the data with respect to a few goalies and teams:

Colorado:

While Aebischer (.909) faced average shot quality and Salo (.917) saw easier shots, it would appear that the defense was lacking when Phil Sauve (.906) was in net, as a higher shot quality was “allowed” against him. This begs the question: why?

Calgary:

It is widely held that Miikka Kiprusoff carried the Flames on his back. However, the Flames saw easier shots than the rest of the league, except when Dany Sabourin played goal (which was only 4 games). However, Kiprusoff exceeded predictions (.933 save percentage vs. .912 predicted) by a significant margin, thus earning the accolades.

Detroit:

Here is an example in the psychology of hockey. Both Dominik Hasek and Curtis Joseph are considered elite goalies. When either of them played, the Red Wings “allowed” tougher shots against—significantly tougher when Hasek was in net (.894 predicted save percentage). The confidence shown in these net minders by their mates was validated, as both played better than anticipated. When the back-up, Manny Legace, played it appears that the Wings tightened up on defense, as the shot quality against became easier (.912). Legace also outperformed predictions.

Florida:

Roberto Luongo was a Vezina candidate and many believe he should have won the award. He did face a lower shot quality than the league average, but it was more difficult than that of Vezina winner Martin Brodeur. Luongo also significantly surpassed predictions, where Brodeur only slightly performed better than predicted.

Minnesota:

The Wild is considered to be a defensive team and that they faced easier shots against should be no surprise. Dwayne Roloson (.910) saw slightly tougher shots than his counterpart, Manny Fernandez (.912), but performed much better (1.03 times) than predicted. In fact, he was tops in the league for save percentage at .933.

New Jersey:

The Devils, like the Wild, are also considered a defensive club. They, too, “allowed” easier shots except when Corey Schwab played, in which case they were average. Martin Brodeur, who won the Vezina trophy, not only faced easier shots than the league average, he only performed slightly better than anticipated (.917 actual vs. .915 predicted).

Pittsburgh:

All Penguins goalies faced tougher than average shots, which may speak to their lack of overall team defense. Of the five goalies used by Pittsburgh, only Aubin and Brochu met or exceeded predictions.

Conclusions

With a well-fit, stable model we were able to calculate predicted shot and save percentages for shots on goal during the 2003-04 season. This leads to a means of classifying the quality of each shot. We then examined each shooter (rolled up at the team level) and each goaltender. From there, we used the model results to devise a shot quality appellation. This allowed us to make inferences about team offense and defense (as it related to various shot qualities) and to assess a team’s shooting percentage as well as a goaltender’s actual vs. predicted save percentage.

Some teams and individuals achieved higher shot quality chances to score than others. It is the *ability to capitalize on these chances* that is key. Furthermore, certain goalies

did, indeed, face tougher shot qualities than others, as evidenced by the model. What is important is how these goaltenders performed *given their circumstances*.

Using this model and methodology of comparison, a team can make accurate player assessments, spot skater and goalie weaknesses as well as make adjustments to any employed system of play.

Actual vs. Predicted Shooting Percentages and Shot Quality

TEAM	Actual			Pred SHOT%	Shot Quality	Index (Actual/Pred)
	SF	GF	SHOT%			
TOTAL	68,806	6,129	8.9	9.1		97.80%

ANA	2,434	180	7.4	8.7	Lower	85.06%
ATL	2,135	208	9.7	9.6	Higher	101.04%
BOS	2,494	203	8.1	8.1	Lower	100.00%
BUF	2,291	210	9.2	8.8	Lower	104.55%
CAR	2,390	168	7.0	9.4	Higher	74.47%
CBJ	2,276	173	7.6	9.9	Higher	76.77%
CGY	2,239	191	8.5	9.0	Lower	94.44%
CHI	2,399	186	7.8	8.5	Lower	91.76%
COL	2,405	230	9.6	9.9	Higher	96.97%
DAL	2,192	189	8.6	8.9	Lower	96.63%
DET	2,475	247	10.0	8.4	Lower	119.05%
EDM	2,327	216	9.3	9.5	Higher	97.89%
FLA	2,270	185	8.2	8.8	Lower	93.18%
LAK	2,343	199	8.5	8.8	Lower	96.59%
MIN	1,977	183	9.3	9.2	Higher	101.09%
MTL	2,255	198	8.8	9.0	Lower	97.78%
NAS	2,223	212	9.5	9.3	Higher	102.15%
NJD	2,428	208	8.6	8.4	Lower	102.38%
NYI	2,329	225	9.7	8.8	Lower	110.23%
NYR	2,409	199	8.3	10.0	Higher	83.00%
OTT	2,409	250	10.4	9.3	Higher	111.83%
PHI	2,401	222	9.2	9.2	Higher	100.00%
PHO	2,160	184	8.5	9.4	Higher	90.43%
PIT	2,027	188	9.3	9.1	Average	102.20%
SJS	2,294	213	9.3	10.1	Higher	92.08%
STL	2,214	182	8.2	9.4	Higher	87.23%
TBL	2,450	235	9.6	9.0	Lower	106.67%
TOR	2,242	231	10.3	9.0	Lower	114.44%
VAN	2,375	231	9.7	9.3	Higher	104.30%
WAS	1,943	183	9.4	9.3	Higher	101.08%

Actual vs. Predicted Save Percentages and Shot Quality

GOALIE	TEAM	Actual			Pred SV%	Shot Quality	Index (Actual/Pred)
		SA	GA	SV%			
TOTAL		68,806	6,129	.911	.909		100.22%
BRYZGALOV	ANA	28	2	.929	.851	Tougher	109.17%
GERBER	ANA	785	64	.918	.906	Tougher	101.32%
GIGUERE	ANA	1,623	140	.914	.909	Average	100.55%
DAFOE	ATL	499	51	.898	.906	Tougher	99.12%
LEHTONEN	ATL	106	5	.953	.894	Tougher	106.60%
NURMINEN	ATL	1,792	173	.903	.906	Tougher	99.67%
POTVIN	BOS	690	67	.903	.907	Tougher	99.56%
RAYCROFT	BOS	1,586	117	.926	.912	Easier	101.54%
BIRON	BUF	1,445	126	.913	.913	Easier	100.00%
MILLER	BUF	73	15	.795	.903	Tougher	88.04%
NORONEN	BUF	818	76	.907	.915	Easier	99.13%
IRBE	CAR	229	23	.900	.908	Tougher	99.12%
STORR	CAR	262	32	.878	.915	Easier	95.96%
WEEKES	CAR	1,651	146	.912	.907	Tougher	100.55%
BRATHWAITE	CBJ	574	59	.897	.908	Tougher	98.79%
DENIS	CBJ	1,970	162	.918	.911	Easier	100.77%
LECLAIRE	CBJ	69	7	.899	.894	Tougher	100.56%
KIPRUSOFF	CGY	966	65	.933	.912	Easier	102.30%
MCLENNAN	CGY	587	53	.910	.911	Easier	99.89%
SABOURIN	CGY	66	10	.848	.905	Tougher	93.70%
TUREK	CGY	463	40	.914	.913	Easier	100.11%
ANDERSON	CHI	601	57	.905	.908	Tougher	99.67%
LEIGHTON	CHI	988	99	.900	.908	Tougher	99.12%
MUNRO	CHI	217	26	.880	.911	Easier	96.60%
PASSMORE	CHI	221	23	.896	.910	Easier	98.46%
THIBAUT	CHI	450	39	.913	.901	Tougher	101.33%
UNDERHILL	CHI	33	4	.879	.913	Easier	96.28%
AEBISCHER	COL	1,703	129	.924	.909	Average	101.65%
SALO	COL	136	12	.912	.917	Easier	99.45%
SAUVE	COL	479	50	.896	.906	Tougher	98.90%
ELLIS	DAL	28	3	.893	.902	Tougher	99.00%
TUGNUTT	DAL	220	22	.900	.906	Tougher	99.34%
TURCO	DAL	1,646	143	.913	.910	Easier	100.33%
HASEK	DET	322	30	.907	.894	Tougher	101.45%
JOSEPH	DET	744	68	.909	.904	Tougher	100.55%
LAMOTHE	DET	58	3	.948	.905	Tougher	104.75%
LEGACE	DET	1,019	82	.920	.912	Easier	100.88%

CONKLIN	EDM	959	84	.912	.914	Easier	99.78%
MARKKANEN	EDM	182	12	.934	.918	Easier	101.74%
SALO	EDM	1,024	107	.896	.911	Easier	98.35%
VALIQUETTE	EDM	7	2	.714	.898	Tougher	79.51%
LUONGO	FLA	2,476	173	.930	.912	Easier	101.97%
SHIELDS	FLA	346	42	.879	.913	Easier	96.28%
CECHMANEK	LAK	1,198	113	.906	.904	Tougher	100.22%
CHOUINARD	LAK	2	0	1.000	.921	Easier	108.58%
HNILICKA	LAK	42	5	.881	.891	Tougher	98.88%
HUET	LAK	961	89	.907	.907	Tougher	100.00%
FERNANDEZ	MIN	1,056	90	.915	.912	Easier	100.33%
ROLOSON	MIN	1,323	89	.933	.910	Easier	102.53%
GARON	MTL	480	38	.921	.918	Easier	100.33%
THEODORE	MTL	1,853	150	.919	.909	Average	101.10%
MASON	NAS	365	27	.926	.910	Easier	101.76%
VOKOUN	NAS	1,958	178	.909	.911	Easier	99.78%
BRODEUR	NJD	1,845	154	.917	.915	Easier	100.22%
CLEMMENSEN	NJD	84	4	.952	.914	Easier	104.16%
SCHWAB	NJD	68	2	.971	.909	Average	106.82%
DIPIETRO	NYI	1,261	112	.911	.912	Easier	99.89%
DUBIELEWICZ	NYI	50	3	.940	.918	Easier	102.40%
SNOW	NYI	932	94	.899	.903	Tougher	99.56%
DUNHAM	NYR	1,522	159	.896	.895	Tougher	100.11%
LABARBERA	NYR	91	16	.824	.891	Tougher	92.48%
MARKKANEN	NYR	614	53	.914	.900	Tougher	101.56%
MCLENNAN	NYR	97	12	.876	.894	Tougher	97.99%
VALIQUETTE	NYR	71	6	.915	.913	Easier	100.22%
EMERY	OTT	52	5	.904	.894	Tougher	101.12%
LALIME	OTT	1,334	127	.905	.908	Tougher	99.67%
PRUSEK	OTT	648	54	.917	.907	Tougher	101.10%
BURKE	PHI	389	35	.910	.901	Tougher	101.00%
ESCHE	PHI	932	79	.915	.919	Easier	99.56%
HACKETT	PHI	684	65	.905	.910	Easier	99.45%
LITTLE	PHI	8	2	.750	.949	Easier	79.03%
NIITTYMAKI	PHI	77	3	.961	.929	Easier	103.44%
BIERK	PHO	108	12	.889	.900	Tougher	98.78%
BOUCHER	PHO	1,145	108	.906	.904	Tougher	100.22%
BURKE	PHO	913	84	.908	.901	Tougher	100.78%
JOHNSON	PHO	243	21	.914	.912	Easier	100.22%
PELLETIER	PHO	84	12	.857	.894	Tougher	95.86%
AUBIN	PIT	574	53	.908	.908	Tougher	100.00%
BROCHU	PIT	19	1	.947	.903	Tougher	104.87%
CARON	PIT	1,179	138	.883	.902	Tougher	97.89%
CHIODO	PIT	260	28	.892	.904	Tougher	98.67%
FLEURY	PIT	675	70	.896	.907	Tougher	98.79%

NABOKOV	SJS	1,610	127	.921	.910	Easier	101.21%
TOSKALA	SJS	760	53	.930	.912	Easier	101.97%
DIVIS	STL	291	29	.900	.892	Tougher	100.90%
JOHNSON	STL	203	20	.901	.901	Tougher	100.00%
OSGOOD	STL	1,604	144	.910	.906	Tougher	100.44%
GRAHAME	TBL	664	58	.913	.912	Easier	100.11%
KHABIBULIN	TBL	1,414	127	.910	.913	Easier	99.67%
BELFOUR	TOR	1,483	122	.918	.910	Easier	100.88%
KIDD	TOR	388	48	.876	.895	Tougher	97.88%
TELLQVIST	TOR	293	31	.894	.909	Average	98.35%
AULD	VAN	168	12	.929	.910	Easier	102.09%
CLOUTIER	VAN	1,554	134	.914	.909	Average	100.55%
HEDBERG	VAN	459	46	.900	.910	Easier	98.90%
CHARPENTIER	WAS	168	21	.875	.900	Tougher	97.22%
KOLZIG	WAS	1,958	180	.908	.912	Easier	99.56%
OUELLET	WAS	210	19	.910	.905	Tougher	100.55%
STANA	WAS	100	11	.890	.912	Easier	97.59%
YEATS	WAS	141	13	.908	.907	Tougher	100.11%

Glossary

- **Rebound Shot** – A rebound was a shot within two seconds of another shot with no intervening “event” with a distance less than 25 feet. [Ryder]
- **Scramble Shot** – A shot less than 6 feet away that was neither a rebound nor an empty net goal. [Ryder]
- **Long Shot** – A shot from more than 61 feet and not an empty net goal. [Ryder]
- **Normal Shot** – A non- rebound, scramble or long shot per above. [Ryder]
- **Kolmogorov-Smirnov (KS) Report** – This is a measure of the model’s effectiveness in rank ordering performance (e.g., goal *versus* save). The KS statistic is calculated by subtracting the cumulative distribution of one performance category (goal) from another (save). The point where maximum separation occurs is the KS spread. It is desirable to maximize this separation – the higher the KS, the more predictive the score. As a scorecard, or model, matures, the KS statistic can be expected to degrade over time. The KS Report will compare the KS statistic from the development (baseline) population to that of another time period (out-of-time validation).
- **Population Stability Index** – The population stability index evaluates changes in the scored population over time. This index measures the separation (divergence) between score distributions from time of development and those from a validation period. Over time, changes in the population may result in corresponding changes in scores. A smaller index value is more indicative of a stable population, suggesting that a validation population is similar to the baseline standard population used in scorecard development.

$$Index = \sum_{i=1}^n \left[\left(\frac{v_i}{V} - \frac{b_i}{B} \right) \times \left(\ln \left\{ \frac{v_i}{V} \div \frac{b_i}{B} \right\} \right) \right] \quad [E. M. Lewis]$$

where v_i is the count of validation records in the i -th score band, V is the total count of the validation records, b_i is the count of baseline records in the i -th score band and B is the total count of baseline records. As a rule of thumb, PSI less than or equal to 0.100 indicate similar populations.

- **Characteristics Analysis** – A characteristics analysis is a comparison of the statistical distribution of counts or percentages of the attributes in the validation population with those in the sample that were originally used to develop the model. Characteristics analysis enables one to identify the cause(s) for the significant changes that a population stability index may detect. From this analysis, the characteristics responsible for the shift in scores can be identified and the contribution to the overall change in the final score can be computed.